

Nucleus Structure & Cell Cycle

Current 71-slide faculty deck primary - official Week 4 objectives integrated - nucleus structure tied directly to proliferation control

5 official objectives

Nuclear transport

Lamins

Cyclin-CDK pairs

Rb restriction point

CDK4/6 inhibition

SOURCE PRIORITY: the uploaded current faculty deck controls terminology, examples, and emphasis. The official Week 4 learning-objective sheet defines what must be mastered. This companion reorganizes the same material for retrieval and application without replacing the source deck.

Official Week 4 Objectives

1. Identify the structural elements of the nucleus, including nuclear membrane, lamins, pores, matrix/scaffold, euchromatin/heterochromatin, and nucleolus, and explain the functional significance of each.
2. Describe the mechanism of nuclear import and export via nuclear pores.
3. Recognize mutations in lamins and other nuclear-envelope proteins that cause laminopathies and nuclear envelopopathies.
4. List major cyclins and CDKs, explain how their expression/activity are regulated during the cell cycle, and describe Rb control of the G1-S transition.
5. Explain how altered cyclin/CDK activation supports cancer-cell growth and the mechanism of CDK4/6 inhibition in cancer treatment.

Chromatin

Euchromatin is lighter/open and transcriptionally accessible. **Heterochromatin** is dense/condensed and less transcriptionally active.

Nuclear envelope

Separates nucleoplasm from cytoplasm. The outer nuclear membrane is continuous with rough endoplasmic reticulum.

Nuclear pores

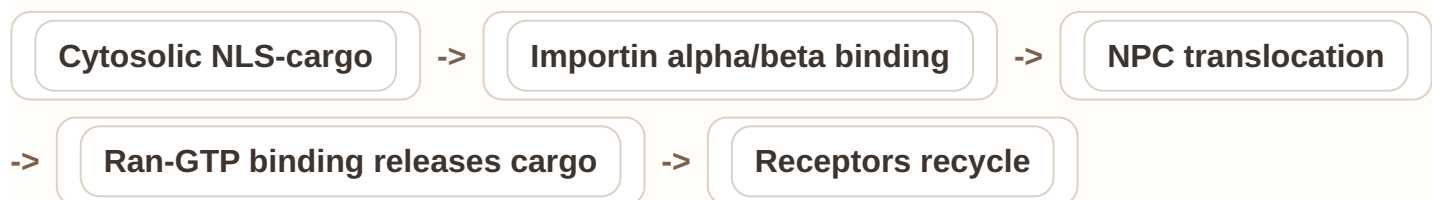
Large protein complexes spanning the envelope and coordinating selective exchange of ions, RNA, proteins, and ribosomal subunits.

Nuclear lamina

Meshwork just beneath the inner nuclear membrane that provides mechanical support and organizes chromatin attachment.

Morphology clue: a small resting lymphocyte is an example of a nucleus with abundant heterochromatin. Proliferating cells tend to have more active nuclear transport and can have increased pore number.

- Small molecules and proteins below roughly 40 kDa can cross more freely by passive diffusion.
- Larger cargo requires a targeting signal, transport receptor, and energy-dependent Ran-GTP cycle.
- **NLS** = nuclear localization signal, usually a short basic amino-acid-rich sequence.
- **NES** = nuclear export signal.
- **Importins and exportins** are karyopherin-family transport receptors.



Directionality comes from the Ran nucleotide gradient. Import/export are not powered by a motor dragging cargo through the pore; they are driven by regulated cargo-receptor interactions tied to Ran-GTP/Ran-GDP.

Nucleolus

Site of rRNA transcription/processing and ribosomal-subunit assembly. The deck emphasizes clustered rRNA genes contributed by 10 chromosomes (5 homologous pairs). 18S, 5.8S, and 28S rRNAs are processed/assembled here.

Multiple nucleoli

Often correlate with active growth/proliferation because high ribosome production is needed for high protein synthesis.

Nuclear matrix / scaffold

A fibrillar protein/RNA network that contributes to nuclear architecture, dynamic chromosome territories, nucleolar organization, and transient DNA attachment sites associated with replication, transcription, and possibly repair.

Slides 28-41

Nuclear lamina, lamin-associated domains, and laminopathies

LO 3

- Lamins A, B, and C are intermediate-filament proteins forming a supportive meshwork beneath the inner nuclear membrane.
- Lamins A and C are encoded by the same **LMNA** gene through alternative splicing.
- Chromatin can attach to the lamina in lamin-associated domains (LADs).

Disorder/idea	Lecture anchor
Laminopathy	Genetic disease caused by lamin defects, especially LMNA-associated disease
Examples	Cardiomyopathy, congenital muscular dystrophy, and progeroid syndromes
Hutchinson-Gilford progeria syndrome	Mutant lamin A product = progerin ; abnormal lamina architecture and defective chromatin interactions
Nuclear envelopathy	Broader category that can involve proteins interacting with lamins, such as emerin

Treatment anchor in the deck: farnesylation inhibition is used for some progeria syndromes; the lecture names Zokinvy as a farnesylation inhibitor.

Slides 42-58

Cyclins and CDKs are the conserved intrinsic cell-cycle control system

LO 4-5

Phase / transition	Major complex to know	Primary job
G1 progression	Cyclin D - CDK4/6	Responds to growth-factor signaling; phosphorylates Rb

G1/S	Cyclin E - CDK2	Commitment to DNA replication
S phase	Cyclin A - CDK2	Supports DNA replication
G2	Cyclin A - CDK1	Prepares cell for mitosis
M phase	Cyclin B - CDK1	MPF; drives mitotic events

How CDKs are regulated

CDKs require cyclin binding and are controlled by phosphorylation/dephosphorylation. Cyclin-dependent kinase inhibitors (CKIs), such as p21 in the lecture, can block CDK activity.

How cyclins are regulated

Cyclin abundance rises and falls. Ubiquitin tagging and proteasomal degradation are central to cyclin turnover and cell-cycle transitions.

M-phase MPF = cyclin B + CDK1. The lecture links MPF to histone H1 phosphorylation/chromosome condensation, lamin phosphorylation/nuclear-envelope breakdown, spindle formation, and fragmentation of Golgi/mitochondrial/ER architecture.

G0 is not cellular inactivity. Quiescent cells are metabolically active but not proliferating. Entry/re-entry depends strongly on growth-factor/hormonal signaling and the cell's differentiation state.

Slides 55-60

The Rb restriction point links growth signals to S-phase entry

LO 4-5

Growth factor

->

Signal-transduction cascade

->

Cyclin D - CDK4/6

->

Rb phosphorylation

->

E2F released

->

G1 to S progression

Active Rb

Unphosphorylated Rb is active. It binds/sequesters E2F and acts as a brake on the G1-S transition.

Inactive Rb

Phosphorylated Rb is inactive. E2F becomes available to drive transcription needed for S-phase entry.

Why CDK4/6 inhibitors work: blocking CDK4/6 prevents efficient Rb phosphorylation. Rb therefore remains active, E2F stays restrained, and the G1-S transition is suppressed. This mechanism only makes sense when the Rb brake is functionally present.

Classic trap: phosphorylation inactivates Rb. Many regulatory proteins become active when phosphorylated, but Rb is the opposite in this pathway.

Master map

High-yield nucleus-to-cell-cycle integration

All LOs

Prompt	Answer anchor
Outer nuclear membrane	Continuous with ER
Large protein import	NLS + importins + Ran-GTP cycle
rRNA synthesis/processing	Nucleolus
Mechanical support beneath inner membrane	Nuclear lamina
LMNA mutant causing premature aging	Progerin / HGPS
Mitotic nuclear-envelope breakdown	CDK1/cyclin B phosphorylation of lamins
G1 growth-factor response	Cyclin D/CDK4/6
Rb state that blocks E2F	Unphosphorylated / active Rb

Exam traps

Protect these relationships

All LOs

Trap 1: Small cargo can diffuse through pores; large macromolecular cargo needs targeting receptors and Ran-dependent transport.

Trap 2: Lamins are intermediate filaments, not microtubules or actin microfilaments.

Trap 3: Cyclins fluctuate through synthesis/degradation; CDK proteins are primarily regulated through binding and phosphorylation state.

Trap 4: Active Rb is the unphosphorylated form.

Trap 5: CDK4/6 inhibition restores/maintains the Rb brake; it does not directly inhibit E2F DNA binding.

30-SECOND FINISH

Envelope connects to ER. Pores use NLS/NES + importin/exportin + Ran. Nucleolus makes ribosomal components. Lamins support the envelope; LMNA disease can produce progerin. Cyclin B/CDK1 drives mitosis. Cyclin D/CDK4/6 phosphorylates and inactivates Rb, releasing E2F for S-phase entry.